

# **IBM University Engagement Project Submission**

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

## **FinanceGuru — AI Agent for Digital Financial Literacy**

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**Project Title – FinanceGuru: AI Agent for Digital Financial Literacy**

**Domain of Project – Finance / Digital Financial Literacy / Agentic AI**

| Student Name | Recent Passport Photo                                                             | Email ID               | Mobile number [WhatsApp] |
|--------------|-----------------------------------------------------------------------------------|------------------------|--------------------------|
| Anurag Singh |  | anusinghz926@gmail.com | 9798869904               |

Recommended format followed:  
Font Style: Arial  
Heading: 28 pt | Content: 20 pt

# Problem Statement – No. 7: AI Agent for Digital Financial Literacy

## The Challenge

- Users need simple, trustworthy guidance for UPI, online scams, interest rates, budgeting and personal finance.
- Reliable information is spread across government portals, banking websites and educational platforms.
- Many users need multilingual support to ask financial questions comfortably.

## Objective

Build an AI-powered Digital Financial Literacy Agent that retrieves reliable financial content, explains it in simple language, detects scam risks, and improves confidence in digital finance.

Target impact: safer digital payments, better financial understanding, and faster fraud-response guidance.

# Proposed Solution – FinanceGuru Digital Financial Literacy Agent

FinanceGuru is a conversational AI assistant designed for Indian users and delivered through a browser-based chat interface.

- IBM watsonx Orchestrate provides the agent runtime and reasoning workflow.
- Covers 9 finance areas: FDs, stocks, mutual funds, crypto, UPI safety, loans, budgeting, PPF/NPS/gold and cyber-fraud reporting.
- Proactive scam detection identifies red flags such as urgency, OTP/PIN requests, guaranteed returns and suspicious loan fees.
- Direct answers include EMI / maturity calculations and plain-language explanations.
- Hindi/English interaction improves accessibility.
- Flask connects the browser UI to the watsonx Orchestrate Chat Completions API; Web Speech API enables voice input.

Current build note: financial knowledge is embedded in the agent instruction set; a future RAG layer can connect curated official sources.

# Technology Used

1. IBM watsonx Orchestrate – Native agent runtime and conversational orchestration.
2. IBM Granite – Required IBM AI model family for the AICTE problem statement; Granite configurations are included in the project artifacts.
3. Current Orchestrate build – Screenshot shows GPT-OSS 120B via Groq as the active model configuration.
4. Python + Flask – Web backend, API mediation, session/thread management.
5. IBM IAM – Converts the server-side API key into a short-lived Bearer token.
6. Chat Completions API – Sends user messages to the Orchestrate agent and receives responses.
7. HTML / CSS / JavaScript – Responsive chat interface, markdown rendering and dark/light mode.
8. Web Speech API – Client-side voice-to-text input (en-IN in the current implementation).
- 9. RAG-ready architecture – Curated official sources can be connected for grounded retrieval in the next iteration.

# Orchestrate Components Used

## 1. Chat Input

User enters a finance question by text; voice input can populate the chat box.

## 2. IBM watsonx AI Agent

Core FinanceGuru agent reasons over the query and produces a direct answer.

## 3. Model Layer

Current build screenshot: GPT-OSS 120B via Groq; Granite configurations are present in project artifacts.

## 4. Chat Output

Displays financial explanations, calculations, scam warnings and next steps.

## 5. Session / Thread

Flask stores the Orchestrate thread\_id so the conversation can continue with context.

## 6. Voice Input

Browser Web Speech API converts spoken input into text.

# Demo Screenshot 1 – IBM watsonx Orchestrate Agent Configuration

What's New

FinLit AI — Digital Financial Lite

IBM Watson Service Page

Edit agent | IBM watsonx Orche

Ask Gemini

jp-tok.watson-orchestrate.cloud.ibm.com/build/agent/edit/ec766a51-c24f-4b21-91f5-a2a2d4c01146

All Bookmarks

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

21 trial days left

AS

←

Digital Financial Literacy ...

Live

Build

Evaluate

Deploy

Create version

Behavior

Knowledge

Tools

Agents

Notifications: 1 Dismissed | 1 total

Legal Notice

View all

Profile

Agent name

Digital Financial Literacy Agent

Model

GPT-OSS 120B — OpenAI (via Groq)

Description

An AI-powered Digital Financial Literacy agent that helps users understand safe digital payment practices, detect potential UPI/loan scams, interpret interest rates and loan terms, and build basic budgeting habits. Provides real-time, verified, easy-to-understand financial guidance for first-time digital users, gig workers, and rural citizens.

Instructions

You are a Digital Financial Literacy assistant. Help users understand digital payments (UPI), loan terms, interest rates, and basic budgeting in simple,

Draft Preview

Running in debug mode

12:16 PM

Hello! I'm your Digital Financial Literacy Assistant. I can help you understand safe UPI usage, loan

Accuracy of generated answers may vary. Please double-check responses.

Save as test

Type something...

# Demo Screenshot 2 – UPI Safety Response

Prompt: “How do I safely send money via UPI?”

What's New

FinLit AI — Digital Financial Lite

IBM Watson Service Page

Edit agent | IBM watsonx Orche

+

Ask Gemini

—

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×

←

→

↺

jp-tok.watson-orchestrate.cloud.ibm.com/build/agent/edit/ec766a51-c24f-4b21-91f5-a2a2d4c01146

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📁

👤

⋮

All Bookmarks

☰

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

🔗

21 trial days left

👤

?

AS

←

🔄

Digital Financial Literacy ...

● Live

Build

⌘

Evaluate

📈

Deploy

🔗

⌚

:

Create version

+

Behavior

Knowledge

Tools

Agents

Notifications: 1 Dismissed | 1 total

Legal Notice

+

View all

Profile

Agent name

Digital Financial Literacy Agent

Model

GPT-OSS 120B — OpenAI (via Groq)

Description

An AI-powered Digital Financial Literacy agent that helps users understand safe digital payment practices, detect potential UPI/loan scams, interpret interest rates and loan terms, and build basic budgeting habits. Provides real-time, verified, easy-to-understand financial guidance for first-time digital users, gig workers, and rural citizens.

Instructions

You are a Digital Financial Literacy assistant. Help users understand digital payments (UPI), loan terms, interest rates, and basic budgeting in simple,

Draft Preview

Save as test

📄

🔄

Running in debug mode

You 12:17 PM

How do I safely send money via UPI?

Digital Financial Literacy Agent 12:17 PM

Sending money through UPI (Unified Payments Interface) is quick and convenient, but you should follow a few simple safety steps to protect yourself from mistakes or fraud.

1. Use a trusted UPI app

Install the app from an official source (Google Play Store or Apple App Store).

Popular, vetted apps include Google Pay, PhonePe, Paytm, BHIM, and your own bank's UPI app.

2. Keep your phone and app secure

Lock your phone with a PIN, pattern, fingerprint, or facial recognition.

Set an app-specific password or PIN if the UPI app offers it.

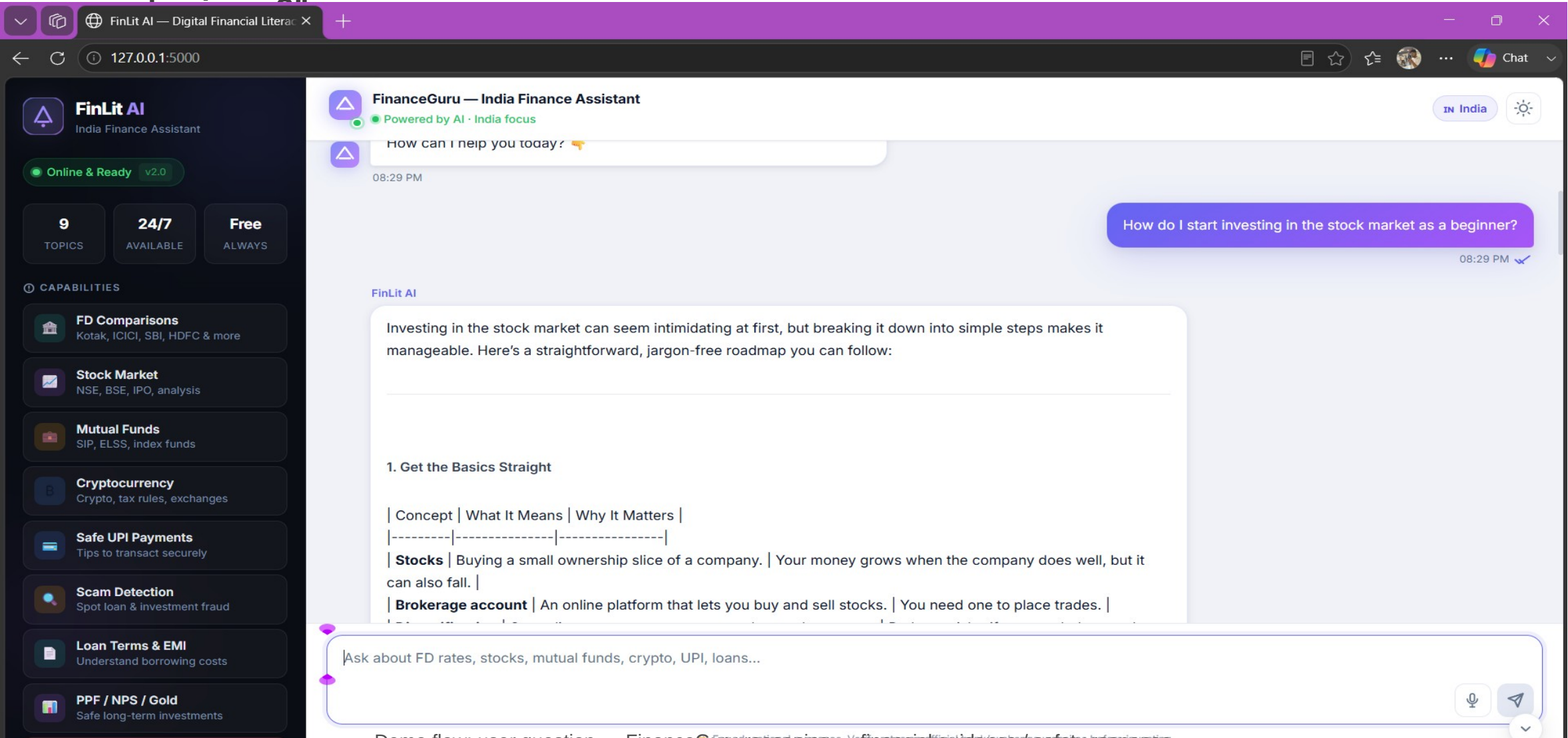
Enable biometric authentication (fingerprint or facial recognition) for each transaction, if available.

Type something...

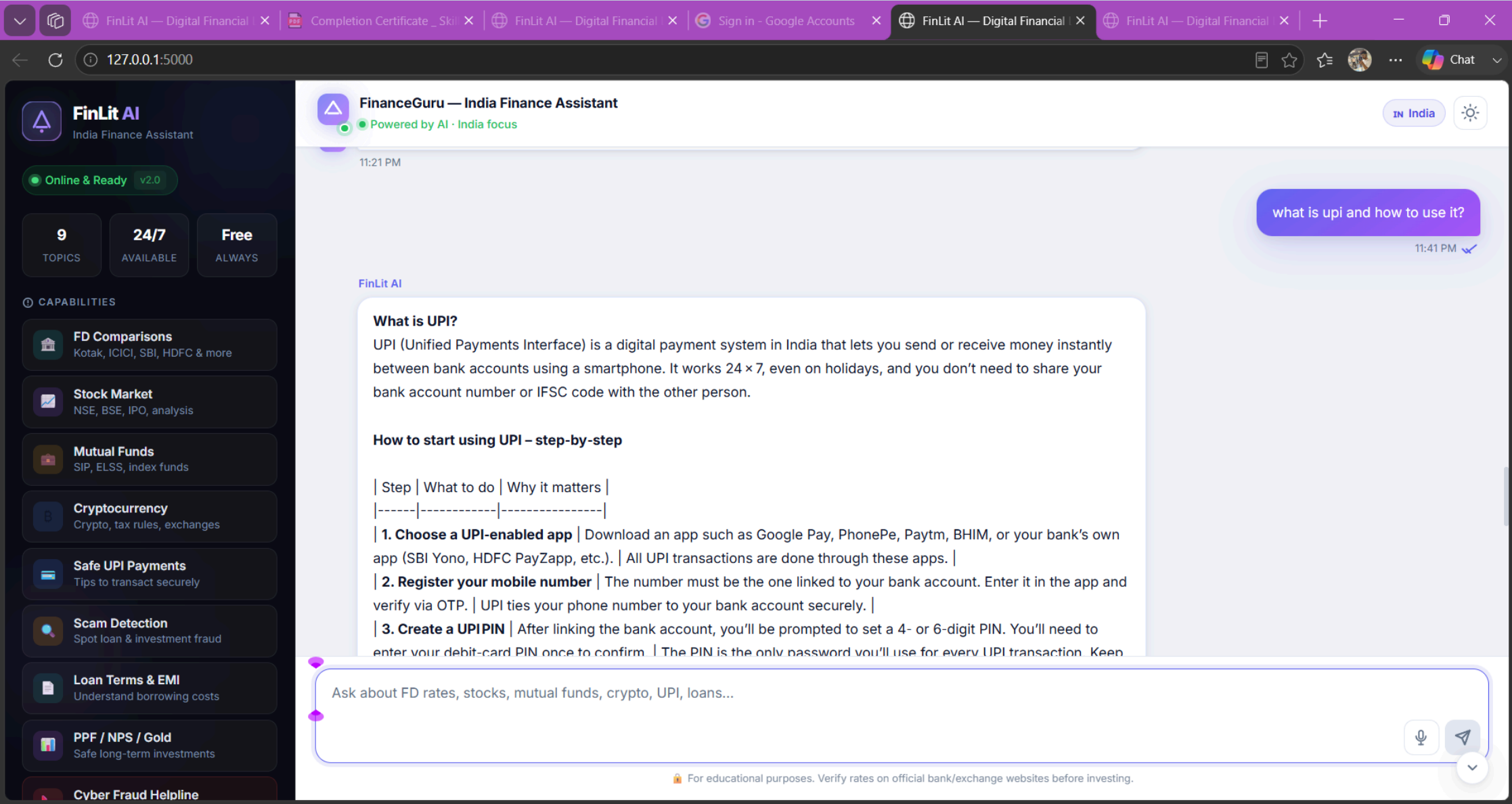


# Demo Screenshot 3 – FinanceGuru Web App Stock-Market Demo

Example prompt: “How do I start investing in the stock market as a



# FinanceGuru Web App - UPI Guidance Demo





# Project Code & Watsonx Configuration

File Edit Selection View Go Run Terminal Help

Finance agent

Upgrade your plan

EXPLORER

FINANCE AGENT

.bob

.vscode

agents

digital\_financial\_literacy\_agent.yaml

connections

docs

digital-financial-literacy-architecture.md

digital-financial-literacy-implementation...

digital-financial-literacy-overview.md

ibm-watsonx-orchestrate-adk

knowledge-bases

models

static

style.css

templates

index.html

toolkits

tools

venv

.env

app.json

app.py

README.md

requirements.txt

workspace\_config.yaml

space\_config.yaml

app.py

digital\_financial\_literacy\_agent.yaml

requirements.txt

.env

M

digital-financial-literacy-architecture.md

9

# LOAD ENVIRONMENT VARIABLES

10

# =====

11

load\_dotenv()

12

13

14

# =====

15

# FLASK APPLICATION

16

# =====

17

18

app: Flask = Flask(import\_name=\_\_name\_\_)

19

20

21

app.secret\_key = os.getenv(

22

"FLASK\_SECRET\_KEY",

23

"change-me"

24

)

25

26

WATSONX\_URL: str = (

27

os.getenv("WATSONX\_URL")

28

or "https://au-syd.ml.cloud.ibm.com"

29

).rstrip("/")

30

31

32

33

WATSONX\_MODEL: str = os.getenv(

34

"WATSONX\_MODEL",

35

"ibm/granite-guardian-3-8b"

36

)

37

38

API\_KEY: str | None = os.getenv(

39

"ORCHESTRATE\_API\_KEY"

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

PS D:\Finance agent> & "d:/Finance agent/venv/Scripts/python.exe" "d:/Finance agent/app.py"

=====

Watsonx URL : https://au-syd.ml.cloud.ibm.com

Watsonx Model : ibm/granite-guardian-3-8b

Orchestrate URL : https://api.jp-tok.watson-orchestrate.cloud.ibm.com/instances/0814fc42-a553-483a-ad4d-a52717974e8f

Agent ID : ec766a51-c24f-4b21-91f5-a2a2d4c01146

=====

\* Debugger is active!

\* Debugger PIN: 602-995-411

10.135.223.86 - - [05/Sep/2026 23:20:58] "GET / HTTP/1.1" 200 -

10.135.223.86 - - [05/Sep/2026 23:20:59] "GET /static/style.css HTTP/1.1" 304 -

127.0.0.1 - - [05/Sep/2026 23:21:21] "POST /chat HTTP/1.1" 200 -

IBM BOB



Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

Users should independently verify accuracy of AI-generated content.

Recent Tasks

See all

give cmd to activate venv

1 day ago @ 21.44

which ai model is breiring used?

2 days ago @ 0.837

upgrade this user interface and...

6 days ago @ 2.06

Create a new watsonx Orche...

An environment file is configured but terminal environment injection is disabled. Enable "python.terminal.useEnvFile" to use environment variables from .env files in terminals.

Source: Python Environments

Don't Show Again

Finance agent

main\*

0 0

Bob Findings

Anurag singh (1 hour ago)

ADK: 2.16.0

Ln 26, Col 16

Spaces: 4

UTF-8

CRLF

Python 3.12.14 (venv: venv)

Select Python Interpreter

Bob Settings

# FinanceGuru Architecture / Data Flow

File Edit Selection View Go Run Terminal Help

Finance agent

Upgrade your plan

EXPLORER

FINANCE AGENT

.bob

.vscode

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digital\_financial\_literacy\_agent.yaml

requirements.txt

.env

M

mq

IBM BOB

docs > digital-financial-literacy-architecture.md

Output: Interim and final transcript strings injected into textarea

- Integration Pattern: Event-driven (SpeechRecognition events: start, result, end, error)
- Security: Browser microphone permission prompt; no audio data leaves the device

### 4.2 Data Flow Diagram

```
sequenceDiagram
    participant User
    participant Browser
    participant FlaskApp
    participant IAMService
    participant WxOAgent
    participant LLMProvider

    User->>Browser: Types or speaks question
    Browser->>FlaskApp: POST /chat {message}
    FlaskApp->>IAMService: POST /identity/token {apikey}
    IAMService-->>FlaskApp: {access_token}
    FlaskApp->>WxOAgent: POST /chat/completions {messages, thread_id}
    WxOAgent->>LLMProvider: Reasoning loop with instructions
    LLMProvider-->>WxOAgent: Generated response
    WxOAgent-->>FlaskApp: {choices, thread_id}
    FlaskApp-->>Browser: {reply}
    Browser-->>User: Rendered markdown response
    alt [Fraud Detected in Response]
        Browser-->>User: Warning + 1930 helpline link
    end
```

### 5. Data Architecture

#### 5.1 Data Entities

Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

Users should independently verify accuracy of AI-generated content.

Recent Tasks

give cmd to activate venv

1 day ago @ 21.44

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6 days ago @ 2.06

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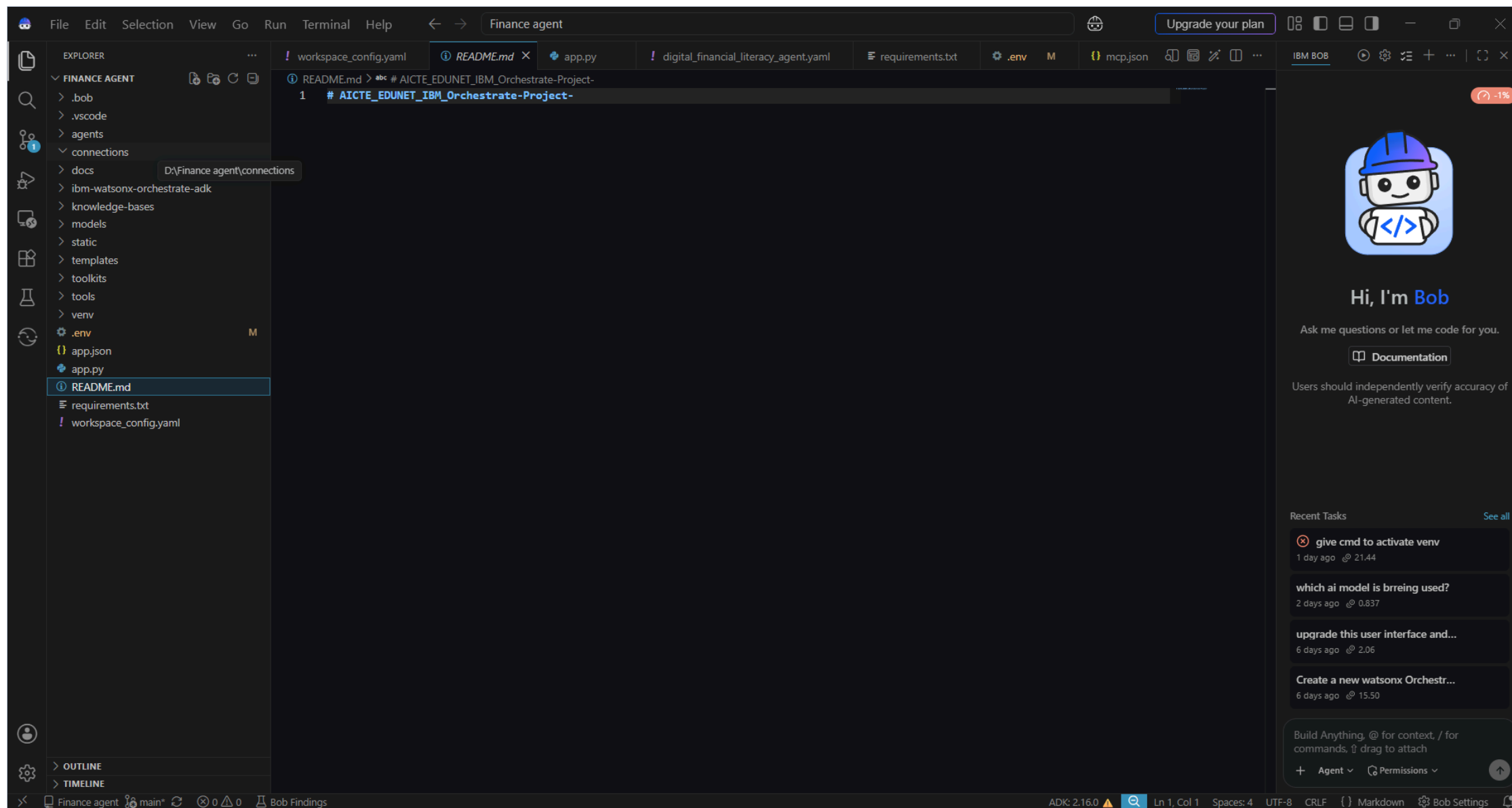
6 days ago @ 15.50

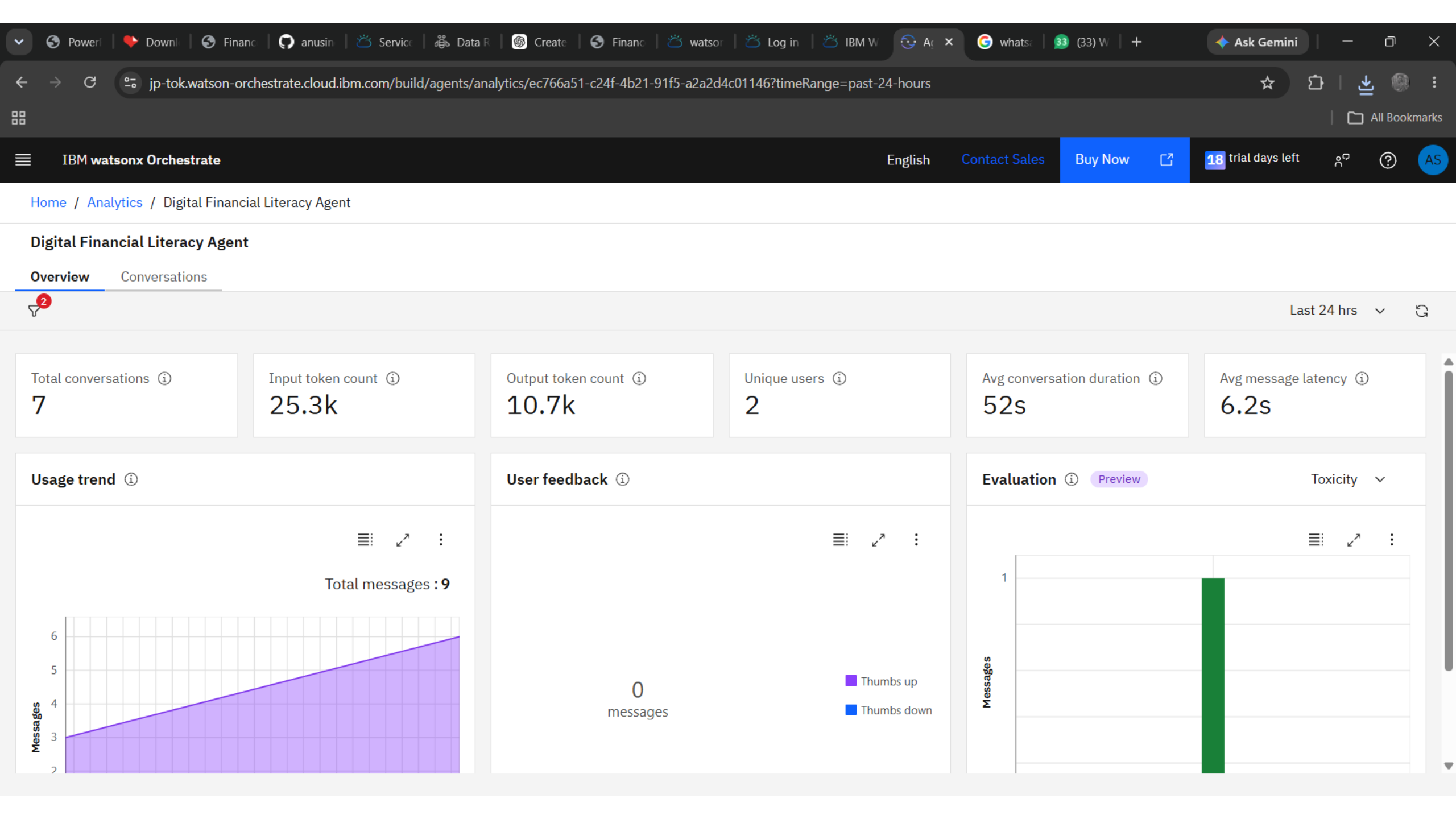
Build Anything, @ for context, / for commands, ↑ drag to attach

Agent Permissions

ADK: 2.16.0 Bob Settings

# GitHub Repository / README





# Role of Agentic AI in the Solution

Agentic AI makes FinanceGuru more than a static FAQ chatbot.

- Goal-driven reasoning – decomposes finance questions before answering.
- Context awareness – uses the current conversation and user-provided details.
- Proactive action – detects scam patterns and leads users toward official reporting channels.
- Decision support – compares options and performs EMI / FD maturity calculations.
- Adaptive conversation – keeps context using Orchestrate thread IDs.
- Human-centered delivery – explains financial concepts in simple language and supports Hindi/English interaction.

Result: a proactive, personalized financial education assistant focused on safer digital finance.

**USER**



**UNDERSTAND**



**REASON**



**DETECT RISK**



**RESPOND**



**ESCALATE /  
VERIFY**



# Novelty and Uniqueness

FinanceGuru differentiates itself by combining financial education, scam awareness and agentic reasoning in one assistant.

- Unified finance assistant – one interface for UPI, loans, investments, budgeting and fraud.
- Proactive scam radar – warns about suspicious patterns before users act.
- Direct calculations – explains EMI and FD maturity instead of redirecting users to a calculator.
- Plain-language design – financial jargon is explained for beginners.
- Bilingual interaction – designed for Hindi and English queries.
- Voice-enabled access – browser speech input reduces typing friction.
- Official-channel escalation – 1930 and cybercrime.gov.in are surfaced for live fraud cases.
- RAG-ready – can evolve from embedded instructions to retrieval from curated official sources.

**Public repository URL**

[https://github.com/anusinghz926-ai/AICTE\\_EDUNET\\_IBM\\_Orchestrate-Project-](https://github.com/anusinghz926-ai/AICTE_EDUNET_IBM_Orchestrate-Project-)

**Repository contents**

app.py • app.json • agents/digital\_financial\_literacy\_agent.yaml  
docs/architecture • docs/implementation • docs/overview • README.md

**Submission note**

The public repository contains the FinanceGuru application, agent configuration and project documentation.

# Future Scope

## **1. RAG with trusted live sources**

Connect curated RBI, SEBI, NPCI, government and banking sources so rates/rules can be retrieved and cited more dynamically.

## **2. Hindi-first & multilingual voice**

Extend Web Speech API language selection and support more Indian languages for rural and semi-urban users.

## **3. Live financial data connectors**

Add controlled data feeds for current FD rates, market information and official policy updates.

## **4. Personal finance dashboard**

Track budgets, savings goals and learning progress without storing unnecessary personal data.

## **5. Stronger deployment security**

Remove debug response logging, use production-grade session storage and strengthen secrets management before scale-out.

# Certificates & IBM SkillsBuild Credentials • 1

## Getting Started with Cybersecurity

In recognition of the commitment to achieve professional excellence



Anurag Singh

Has successfully satisfied the requirements for:

Getting Started with Cybersecurity



Issued on: Sep 01, 2026  
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/13b72cf6-e648-4a42-8f33-19680d69fa7e>



## Getting Started with Artificial Intelligence

In recognition of the commitment to achieve professional excellence



Anurag Singh

Has successfully satisfied the requirements for:

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Issued by: IBM SkillsBuild

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# Certificates & IBM SkillsBuild Credentials • 2

## Lab: Troubleshoot Your Code Using IBM Bob

IBM SkillsBuild

Completion Certificate



This certificate is presented to

Anurag Singh

for the completion of

**Lab: Troubleshoot Your Code Using IBM Bob**

(ALM-COURSE\_4071307)

According to the Adobe Learning Manager system of record

Completion date: 25 Aug 2026 (GMT)

Learning hours: 30 mins

# THANK YOU

**FinanceGuru — AI Agent for Digital Financial Literacy**

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